

```
!pip install matplotlib seaborn pandas numpy
```

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Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post1)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

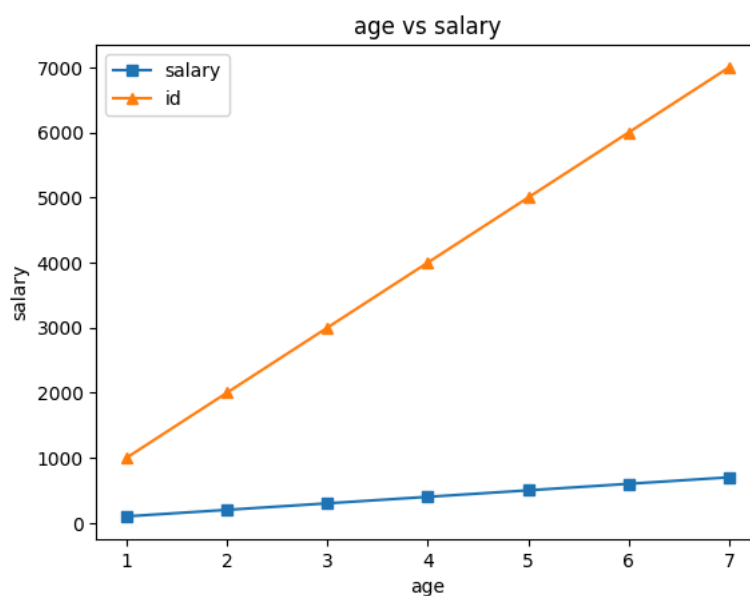
```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
```

```
age=[1,2,3,4,5,6,7]
salary=[100,200,300,400,500,600,700]
id=[1000,2000,3000,4000,5000,6000,7000]
```

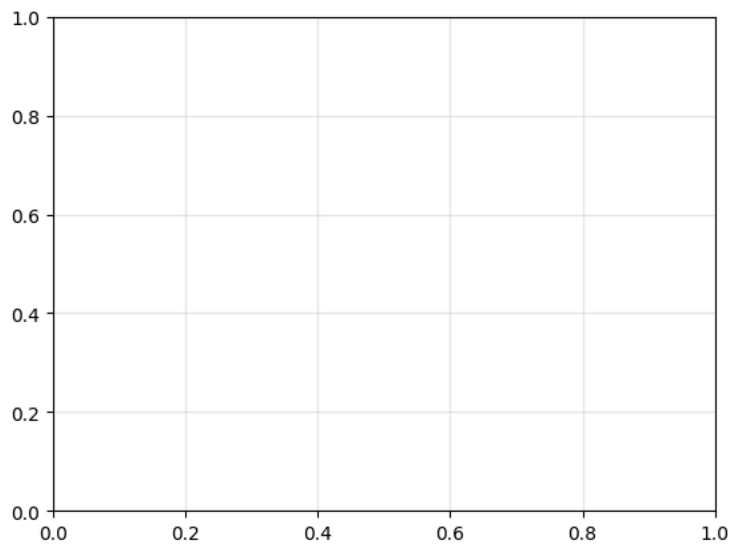
```
plt.figure(figsize=(20,2))
```

```
<Figure size 2000x200 with 0 Axes>
<Figure size 2000x200 with 0 Axes>
```

```
plt.plot(age,salary,marker="s",label="salary")
plt.plot(age,id, marker="^" , label="id") # Changed marker from '^' to '^'
plt.legend();
plt.title("age vs salary")
plt.xlabel("age")
plt.ylabel("salary")
plt.savefig("age_vs_salary.png")
plt.show()
```

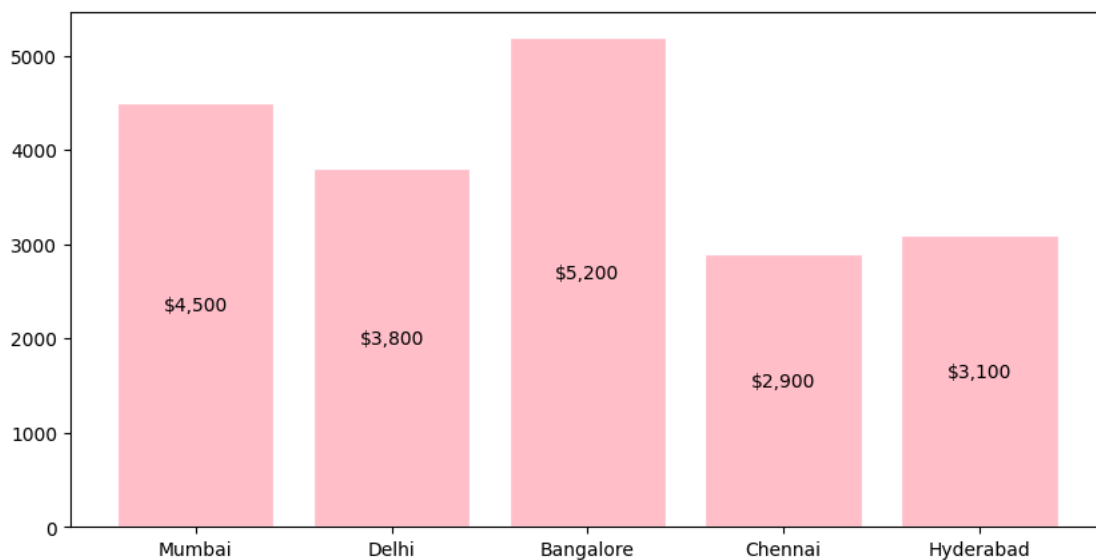


```
plt.grid(True, alpha=0.3)
```



```
cities=['Mumbai','Delhi','Bangalore','Chennai','Hyderabad']
revenue=[4500,3800,5200,2900,3100]
fig,ax=plt.subplots(figsize=(10,5))
bars=ax.bar(cities,revenue,color="pink",edgecolor="white")

for bar in bars:
    height=bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2,height/2,f'${height:,.0f}', ha='center',va='bottom')
```



```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
```

```
sns.set_theme(style='whitegrid')
df=sns.load_dataset('titanic')
print(df.head())
```

```
survived  pclass    sex  age  sibsp  parch    fare embarked  class \
0         0        3  male  22.0     1     0   7.2500        S  Third
1         1        1  female  38.0     1     0  71.2833        C  First
2         1        3  female  26.0     0     0   7.9250        S  Third
3         1        1  female  35.0     1     0  53.1000        S  First
4         0        3  male   35.0     0     0   8.0500        S  Third

   who  adult_male  deck  embark_town  alive  alone
0  man         True  NaN  Southampton    no  False
1 woman        False   C   Cherbourg   yes  False
2 woman        False  NaN  Southampton   yes   True
3 woman        False   C   Southampton   yes  False
4  man         True  NaN  Southampton    no   True
```

